Milestone 3

ECE 4950

Group 18

10/22/2026

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Executive Summary

Milestone 3 focused on developing and integrating the control subsystem for Group 18's automated washer positioning system. The primary objective was to establish communication between the physical motor hardware, MATLAB, and Simulink while developing a graphical user interface (GUI) for identifying and selecting washers on the game board. A hardware in the loop (HIL) approach was used to evaluate motor behavior and support the development of a closed loop position control.

The software integrates camera image acquisition, image processing, washer identification, and motor control. By comparing background and current images, the system identifies washers and stores their position for selection through the GUI. The selected washer's position is used to calculate the required motor movement, which is communicated through Simulink to the Arduino controlled motor. Motor control experiments also examined the effects of proportional, integral, and derivative (PID) controller parameters on motor response, providing a basis for further tuning and performance evaluation.

Additional milestone activities included financial analysis, employee training, and safety assessment. The financial analysis estimated operating expenses and overhead costs, while the training program outlined activities to develop engineering competencies. A Design Failure Mode and Effects Analysis (DFMEA) identified potential mechanical, electrical, and control hazards and documented implemented safety measures. Future development will focus on validating and integrating the washer retrieval mechanism.



Control Subsystem Design

Overview of Hardware-in-the-Loop

Hardware in the Loop (HIL) is a testing and development method that integrates physical hardware with a simulated or virtual controlled environment. It allows engineers to evaluate the behavior of control systems under realistic operating conditions without requiring the entire system to be physically implemented. In this project, HIL is used to investigate the position control of a DC motor through a real time workstation. The motor serves as a physical actuator, while the workstation executes the control algorithm, processes feedback and adjusts the motor's response to a specified reference position.

The primary difference between a full software simulation and HIL is the involvement of physical components. In a full simulation, the motor, controller, and system dynamics are represented through mathematical models. In HIL, physical hardware participates in the control loop, allowing engineers to observe actual responses that may differ from idealized simulation results. These differences can include electrical noise, mechanical friction, communication delays, and variation in motor performance.

The primary advantages of HIL include realistic performance evaluation, early identification of control system limitations, and the ability to adjust controller parameters using measurable feedback. For this project, HIL provides a practical approach to evaluating motor positioning accuracy, system stability, and the effects of proportional, integral, and derivative gains. These experiments support the development of a reliable, closed loop monitor control system capable of positioning a pointer toward selected locations on the game board.

Document Hardware

The washer positioning system uses a 12 V DC motor controlled by an Arduino Mega through an L298N motor driver. The Arduino serves as the interface between the Simulink control model and the physical hardware, allowing motor commands to be executed based on the desired angular position. The L298N driver provides the necessary interface for controlling motor direction and supplying power to the motor, since the Arduino cannot drive the motor directly. The L298N motor driver supports two DC motors, although only one channel is required for the current design.

Arduino digital pins 4 and 5 are connected to the driver's IN1 and IN2 inputs, respectively. These inputs determine the direction of motor rotation. Setting IN1 HIGH and IN2 LOW produces clockwise rotation, while reversing these states produces counterclockwise rotation. The driver uses a separate motor power supply, allowing the motor to operate at a voltage appropriate for the application. Although the motor is rated for 12 V, lower voltage operation may provide sufficient movement for the current prototype, subject to testing under the required mechanical load.

This configuration establishes the hardware interface between the Arduino, motor driver, and DC motor. The wiring arrangement is illustrated in Figure 1, while the MATLAB and Simulink control system provides the positioning commands used to operate the mechanism.

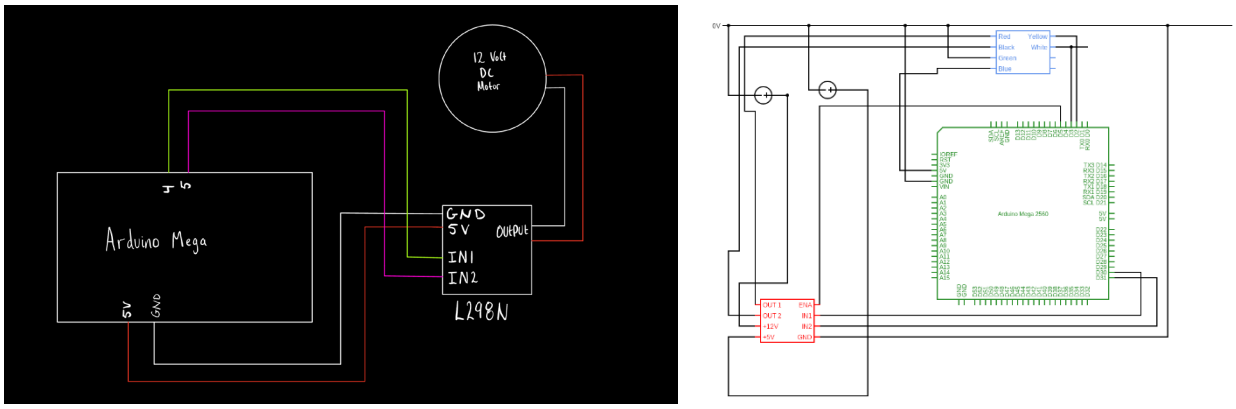


Figure 1. Wiring Diagrams

The integration of the Arduino Mega and the L298N motor driver allows motor operation to be controlled through the Simulink model rather than through manually programmed movement sequences. This configuration provides flexibility for testing different control parameters without requiring significant modifications to the physical hardware. The use of a single motor channel also leaves the second channel of the L298N available for potential future applications, although additional hardware requirements would need to be evaluated before expansion.

An important consideration is the relationship between the motor supply voltage, available torque, and rotational speed. While the motor is rated for 12 V operation, the prototype may operate at a lower supply voltage if sufficient torque is available to move the positioning mechanism. However, the L298N introduces a voltage drop that reduces the voltage delivered to the motor. Consequently, the selected operating voltage should be verified through physical testing. Additional testing should evaluate the motor response under load, positioning accuracy, and the consistency of movement in both rotational directions. These observations will help determine whether the existing hardware configuration provides sufficient performance for the completed washer positioning system.

Document Software

The software was developed to provide an interface between the computer vision system, the user, and the motor positioning hardware. Its primary objective is to identify colored washers on the game board, allow the user to select a desired washer, and generate the motor positioning command required to point toward its location. The system was originally developed using MATLAB scripts (.m) and subsequently integrated into a MATLAB App Designer application (.mlapp). This transition combines image acquisition, object detection, washer selection, and motor control into a single (GUI).

The software operates through four primary stages: initialization, background acquisition, current image acquisition, and image processing. During initialization, the application establishes communication with the webcam and loads the Simulink motor control model. The user then captures a background image of the empty game board, followed by a current image containing the washers. Background acquisition was intentionally maintained as a manual operation during the milestone to allow the team to investigate how variations in the background influence object detection.

The image processing stage compares the background and current images to identify objects introduced onto the game board. The resulting detections are processed to distinguish individual washers and determine their locations. Each detected washer is assigned to a unique identifier and corresponding color label. The software organizes this information within a data structure named gameState, which maintains the information required for washer identification, display, and subsequent selection. The software flowchart in Figure 3 illustrates this sequence and identifies the image and object information associated with the data structure.

Following image processing, the detected washers are displayed in the GUI and made available through a selection menu. The user selects the desired washer, and the application uses its stored position to determine the required motor movement. The resulting position command is transferred to the Simulink control model, which communicates with the Arduino Mega and associated motor driver to operate the DC motor. This arrangement separates visual object identification from physical motor control while allowing both operations to be coordinated through the MATLAB application.

The software also supports repeated image acquisition, allowing the user to process a new arrangement of washers without restarting the entire application. This provides a foundation for further development of the automated positioning system. Future improvements will focus on implementing sequential washer assignments, expanding the interface to represent the six game board wells, and validating the accuracy of the complete vision and motor control workflow.

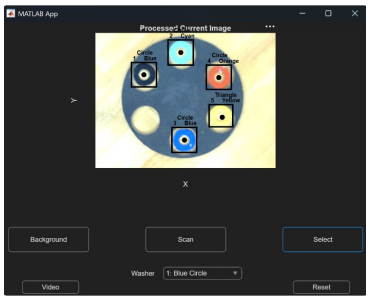


Figure 2. Milestone 3 GUI

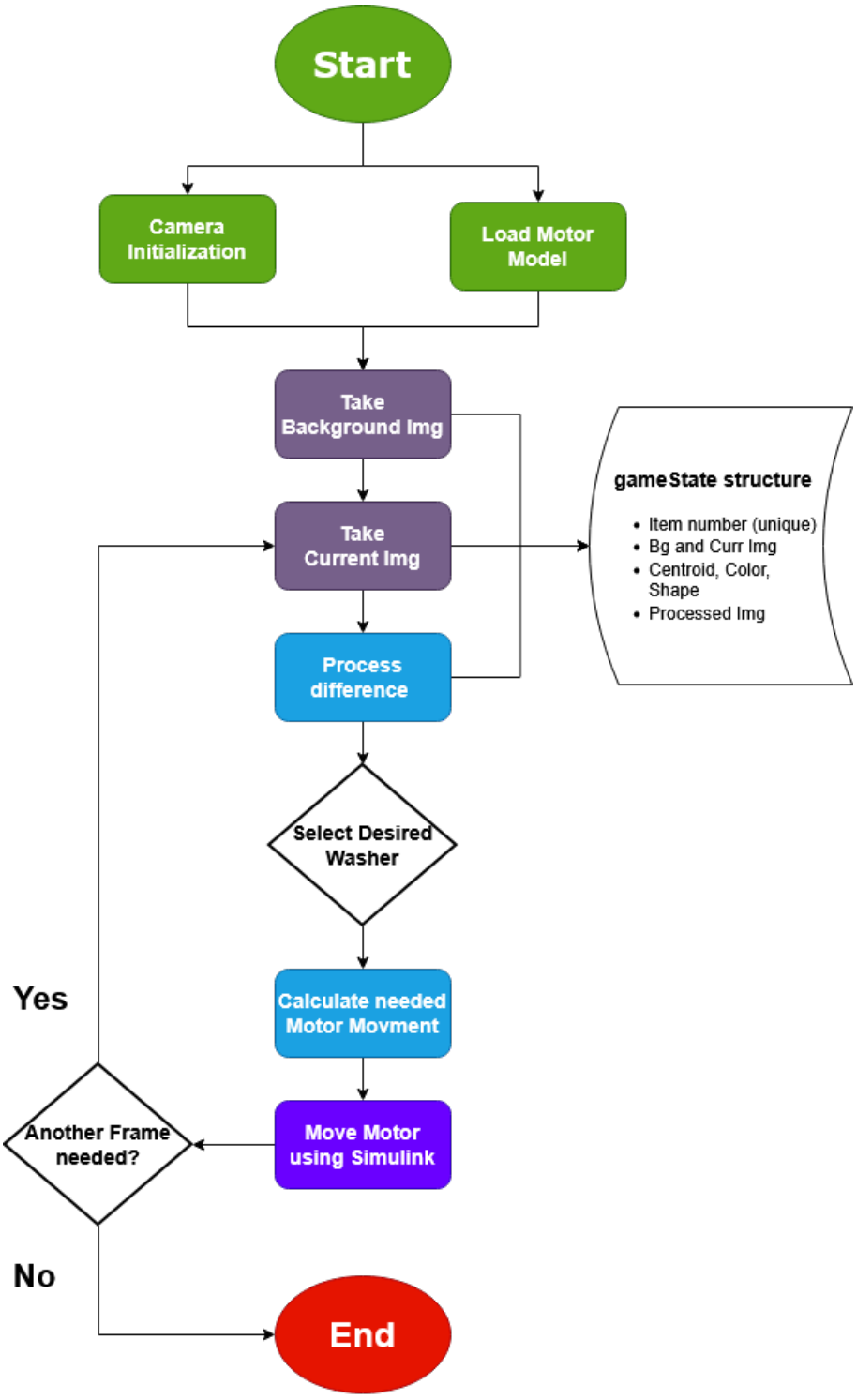


Figure 3. Software flowchart

Document Motor Experiments

The motor experiments focused on evaluating the interaction between the MATLAB software, Simulink control model, Arduino Mega, and DC motor. Initial testing involved establishing communication between the software and physical hardware using the wiring diagrams and pin assignments specified in the Simulink mode. Once the connections were established, the proportional, integra, and derivative (PID) controller parameters were adjusted to investigate their effects on motor response, positioning accuracy, and stability.

Experimental observations indicated that increasing the proportional gain (P) produced a more aggressive response to positional error, resulting in increased motor movement, overshoot, and noticeable oscillations around the desired position. The derivative gain (D) was used to reduce these oscillations by responding to changes in positional error, providing additional damping and improving the smoothness of motor movement. However, excessive derivative action can also reduce responsiveness and amplify measured noise. To evaluate the system under changing reference positions, the Simulink model was configured to generate a sinusoidal reference signal ranging from negative 180° to positive 180°. This allowed the team to observe motor behavior across multiple commanded angular positions.

Following experimental adjustments, the controller parameters were selected as (P = 0.7, I = 0, and D = 0.05). These settings were chosen based on the observed motor response during testing. The corresponding Simulink plots illustrate the motor's positional response and error signal throughout the experiment. Although the motor demonstrated the ability to respond to changing position commands, the results also indicate transient oscillations and error spike during portions of the test. Further experimentation is necessary to quantify overshoot, settling time, and steady state positioning error before the controller's performance can be fully evaluated.

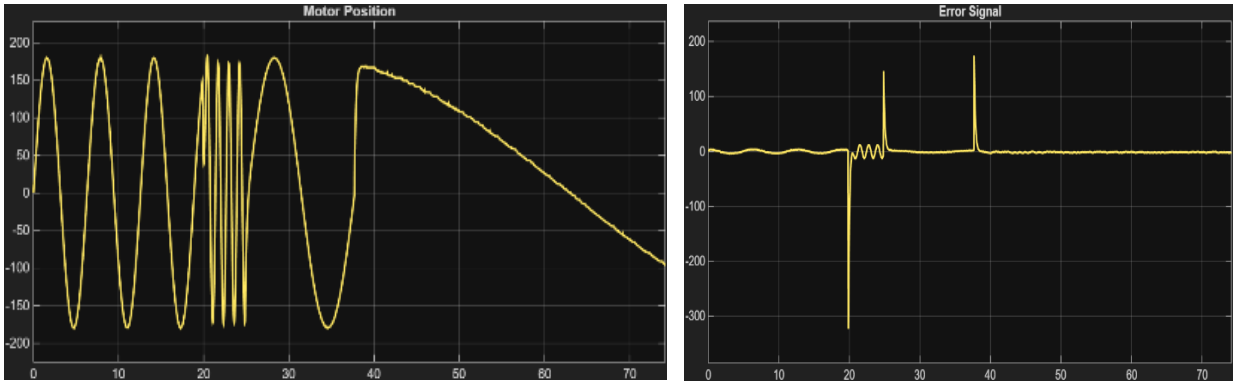


Figure 4. Motor Position and Error plots

The motor position and error plots provide additional insight into the performance of the control system. The error signal remained relatively small during potions of the experimented but exhibited significant spikes during rapid changes in the commanded position. These results suggest that the controller requires further tuning to improve its response to sudden position changes. Future testing should focus on reducing transient error and oscillations while maintaining sufficient responsiveness. Comparing the commanded and measured motor positions will also help quantify tracking accuracy and identify areas for improvement. This analysis will support the development of more consistent and reliable motor positioning.

Document and Evaluate your User Interface

The graphical user interface provides a straightforward method for interacting with the washer positioning system. Its controls separate image acquisition, washer detection, object selection and motor positioning into distinct operations. The processed image displays identified washers, while the automatically populated dropdown allows users to select a target without manually entering its coordinates. This design reduces the number of required user inputs and supports the existing motor control workflow.

One potential improvement would be to incorporate a configuration interface representing the six numbered wells on the game board. Each well could have an individual dropdown menu allowing users to assign a washer from those identified during the initial scan. The available selections would be limited to the detected washers, preventing users from assigning nonexistent washers or selecting the same washer for multiple wells. Once all desired assignments are configured, the system could execute the required movements sequentially without additional user input. This approach would simplify the selection process, provide a clear representation of the intended washer arrangement, and improve overall operating efficiency. Implementing this functionality would require additional assignment validation and movement sequencing logic within the existing GUI and motor control system.

To ensure ease of use, the GUI was kept simplistic. There is a center display that will update to show the current video feed, background image, current image, and processed images. For the controls, there are 5 buttons and a dropdown menu:

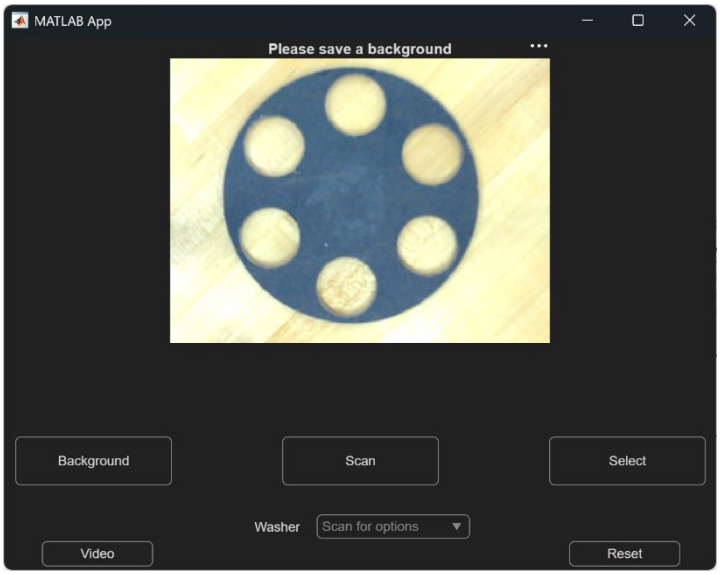


Figure 5. Milestone 3 GUI

- Background:
  Captures the current live frame and stores it as a background image
  Game board must be clear.
  (Only needed once)
- Scan:
  Captures the current live frame and processes the foreground
  Auto populates the Washer drop down menu
- Washer:
  Select desired washer
- Select:
  Move motor to point at the desired washer
- Video:
  Brings up live video feed
- Reset:
  Clears stored values and restarts video feed

Financial Analysis

Personnel	5 Engineers @ $55K/yr + President @ $75K/yr + Admin. Asst. @ $25K/yr = $ 375,000
Fringe Benefit (FB)	A fringe benefit is a form of pay for the performance of services. (5 x $55,000 + $75,000 + $25,000) x 0.36 = $ 135,000
Building	Initially rent a suite of offices with 2 engineers/office (12' x 14'), an office/conference room for President (12' x 20'), and a reception/office area of 16' x 20'. (3 cubicles) x (12' x 14'/cubicle) + President office of (12'x 20') + Reception/office area of (16' x 20') = 1,064 sq ft. Use nominal figure for office space in industrial park sectors of` Clemson area, $9.50/sq ft/mo. Then the lease rate for office space will be $0.79/sq ft/mo x 1,064 sq ft = $841 /mo. = $10,087 /yr.
Furniture	Rental of a desk, chair, and credenza set will run for about $60/mo. Need 7 sets for a total monthly expenditure of $420/mo = $ 5,040/yr. The remaining equipment, furniture and software expenses are estimated to be about 7 computers @ $1500/computer $10,500 7 sets of general software @ $1000/set $7,000 Specialized software $18,000 Copier, printer $4,000 Table and chairs for conference room $3,888 7 telephones @ $35/ea $245 Total $48,673
Phone and Internet	According to Bell South, the cost of a combined voice/data line, is $70.00/mo for operation for 7 telephones the total cost will be $5,880 /year. Assume that long distance calls add another 40% to this to get a total estimated annual phone cost of $8,232
Travel	Another cost item which will be important is travel. There will have to be continual contact with potential clients, attendance at selected technical conferences and workshops, and visits to plants or other locations where potential clients might be. Assume (modestly) that this will that the cost per local trip is $200 and the cost per out-of-state trip is $3,000 there will be 2 of each trip each month $6,400/mo for the first year, or an annual total of $76,800.
Interest	Capital (i.e. money) is needed to fund these initial purchases as well as to underwrite operating expenses until a revenue stream is established by selling engineering services to customers. Assume that through personal contacts a credit line of $800,000 has been established. This is to be repaid over the period of a year with 11 equal payments starting 1 month after the loan date. The negotiated interest rate is 5% per year. The monthly payment M is calculated from $$M = P\frac{(I/q)(1 + I/q)^q}{(1 + I/q)^q - 1} = \$74{,}726$$ Where P is the principal amount ($800,000), I is the interest rate (5%), and q is the number of payments to be made (11). From this, Debt Service = Total interest paid in year = 11 x M - P = $21,983.

Cost Estimate

Salaries	$375,000	FB @ 36%	$135,000
Building	$10,087	Furniture	$48,673
Debt service	$21,983	Travel	$76,800
Internet and Phone Service	$8,232	**Total Costs:**	$675,775

Overhead Calculation

Now we will estimate the Overhead (Indirect Technical Expense) we must charge to recover our costs. This cannot be too large, or else we will price ourselves out of business. On the other hand, we must be realistic, or else we will go broke, and therefore out of business.

Assume that the first year, the 5 engineers will be at least 75% "sold", i.e., 75% of their total time can be charged to customers. Then we can bill

5 engineers @ 75% sold	$206,250	(salaries billable to clients)
FB @ 36%	$74,250	(FB billable to clients)
Total Billable to Clients	$280,500	

The remaining salary dollars and FB's must be charged to overhead.

Total Expenses = Total Costs - Total Billable to Clients = $395,275 (Overhead Number)

This implies an Overhead rate of

OH rate = ($395,275/$280,500) x 100% = 140.92%

= (Overhead Number / Total Billable to Clients) x 100%

This implies that every labor dollar (at the "loaded" rate, i.e. with FB's) must be increased by a factor of

2 (1+ (OH rate/100%) + (5% profit/100%)) in order to recover the costs of doing business and make a profit (assuming a 5% profit). This is the figure that you will use when estimating the cost of a contract to a customer in a proposal. An overhead rate of 150% means that for each $1.00 of direct labor budgeted for a project; $1.50 needs to be budgeted for overhead costs.

Using the Overhead Number

You estimate that a project will take 1 week (40 hours) of your time, i.e. what does it cost for one week of an engineer's time. How much do you bill your client for this time?

Bill to Client

$$= \left[\frac{1 \text{ week work}}{52 \text{ weeks per year}} \bullet \left(\frac{\text{salary} = \$55K}{\text{year}} + \frac{\text{FB} = 0.36 \square \$55K}{\text{year}} \right) \bullet \left(1 + \frac{\text{overhead rate}}{100\%} + \frac{\text{profit} = 5\%}{100\%} \right) \right]$$

= $ 47,304.24

Employee Training Program

Group 18 proposes a yearlong training program for five engineering roles in its Clemson startup. Each employee completes two activities to maintain existing skills and develop new competencies in control systems, embedded firmware, computer vision, hardware safety, and project delivery. Employees document what they learn and apply it to engineering work.

Role, activity, schedule	Training benefit	Time and estimated cost
Controls engineer Q1: Motor control and PID tuning lab	Tune PID gains and record motor overshoot, settling time, and position error.	12 hours $1,311.28
Controls engineer Q3: PLC and industrial control workshop	Program sensor interlocks, stop logic, and fault recovery in a PLC exercise.	12 hours $1,161.28
Embedded system engineer Q1: Embedded C and device debugging course.	Use debugging and unit tests to improve embedded firmware reliability.	12 hours $1,261.28
Embedded system engineer Q3: Real time communication lab	Check communication latency, real time testing, and actuator data exchange.	12 hours $1,161.28
Computer vision engineer Q2: Cameral calibration training	Calibrate camera position and quantify game board measurement error.	12 hours $1,211.28
Computer vision engineer Q4: Object detection and validation lab	Test object detection under changing lighting and document false detections.	12 hours $1,161.28
Hardware integration engineer Q2: Electrical wiring and safety workshop	Practice wiring inspection, strain relief, insulation, and safe cable routing.	12 hours $1,211.28
Hardware integration engineer Q4: Prototype reliability testing lab	Inspect arch fasteners, moving parts, and actuator connections for defects.	12 hours $1,211.28
Technical project manager Q2: Project scheduling and budgeting course	Prepare project schedules, budgets, milestones, and resource assignments.	12 hours $1,361.28
Technical project manager Q4: DFMEA and design review workshop	Use the DFMEA matrix to prioritize risks and document corrective actions.	12 hours $1,161.28

Table 1. Annual Employee Training Program: Activities, Benefits, Schedule, and Estimated Costs

Annual training investment and rationale. Each of the five employees will complete 24 training hours per year, totaling 120 hours. At the financial analysis billing rate of $88.44 per hour, the opportunity cost of employee time is $10,612.80. Including $1,550 in estimated registration and materials, the annual budget is $12,162.80, or $2,432.56 per employee. This allocation maintains skills while reducing rework, improving electrical safety, strengthening system reliability, and supporting project delivery. Each activity produces a demonstration, test record, risk review, or project plan. Training outcomes are reviewed quarterly.

Safety Analysis

The purpose of this analysis is to identify potential hazards in the motorized washer positioning system and evaluate the design measures taken to reduce risk to the user. The system includes a motor driven mechanism, an electromagnet, a solenoid, a camera, a supporting arch, control electronics, wiring, and status LEDs. A Design Failure Mode and Effects Analysis (DFMEA) was developed using the component, failure mode, symptom, effect, probability, severity, and risk index approach presented in the ECE 4950 engineering design safety lecture [1].

The probability rankings are A (very probable), B (probable), C (occasional), D (remote), and E (improbable). Severity rankings are I (catastrophic), II (critical), III (marginal), and IV (negligible). The ratings below are preliminary engineering judgments rather than measured failure frequencies. They describe credible hazards and provide a basis for prioritizing safeguards.

Component	Failure mode	Symptom	Potential effect	Prob	Severity	Index
Electromagnet and washer	Abrupt release or movement	Washer moves unexpectedly	Impact or pinch injury	C	II	II C
Motor and moving arm	Excessive speed	Rapid or uncontrolled motion	Contact injury or equipment damage	C	III	III C
Arch and mounted devices	Loose support or mount	Camera or solenoid shifts	Failing component or loss of alignment	D	III	III D
Electrical wiring	Exposed or obstructive cables	Wire snagging disconnection, or short	Trip, shock, or equipment damage	C	II	II C
User interface	Unclear operating state	User approaches active mechanism	Unexpected motion during handling	C	III	III C
Control System	Incorrect command or loss of feedback	Mechanism does not stop at target	Unintended motion or collision	D	II	II D

Table 2. Preliminary DFMEA for the motorized washer positioning system

The highest priority identified hazards are unintended washer movement and electrical wiring faults, each assigned an initial risk index of II C. The control system also presents a potentially critical consequence if positioning commands or feedback fail. These rankings are conservative screening estimates and should be reviewed after physical tests.

The lecture risk assessment matrix combines probability and severity to categorize risk and determine where design intervention is needed [1]. The following controls were reported as implemented by Group 18. Their intended effects are documented here; quantitative evidence that they reduce risk to an acceptable level has not yet been provided.

Hazard	Implemented control	Safety purpose
Unexpected washer motion	A solenoid was integrated into the control mechanism to prevent the electromagnet from abruptly propelling the washer upward or downward when switched on or off.	Reduces the opportunity for sudden washer motion and impact.
Excessive motor speed	The motion speed of the mechanism was limited.	Reduces the energy and severity of foreseeable contact with moving parts.
Loose or exposed devices	A rigid arch was installed across the platform and mounted on both sides. The camera and solenoid were attached to the arch.	Supports component positioning and reduces exposure to loosely placed hardware.
Wiring interference	Electrical wiring was routed and managed away from the end user.	Reduces snagging, accidental disconnection, and unintended contact with cables.
Unclear system state	LED indicators were included to distinguish start and stop states.	Provides visible operating state information before a user interacts with the mechanism.

Table 3. Implemented safety measures and their intended risk reduction.

The implemented controls address major mechanical, structural, electrical, and human interface hazards. Nevertheless, an LED indicates a commanded state and does not independently prove that the motor has stopped, similarly, the solenoid must be evaluated during repeated electromagnet transitions to confirm that the washer remains restrained. The arch and fasteners should be checked for stability, the managed wires inspected for damaged insulation or pinch points, and the speed limit verified during the normal and abnormal commands.

The final risk assessment should be completed after verifying these protections on the assembled system. Useful acceptance checks include repeated start and stop operations, observation of washer retention, controlled motor response, physical stability of the mounted arch, and correct indication of system states. If any unexpected motion, electrical defect, or structural instability is observed, the affected function should not be used until the hazard is corrected.

Group 18 incorporated safety into the design by addressing washer retention, limiting mechanism speed, securing the camera and solenoid to a platform mounted arch, managing wiring away from the user, and adding visible start and stop indicators. These modifications reduce identifiable hazards and demonstrate that DFMEA findings informed the design. However, without documented verification results, the system cannot yet be conclusively classified as safe for customer operation.

References

[1] ECE 4950, “Engineering Design Safety,” course lecture slides, Clemson University, 2026.

ECE 4950 Project 3 –Closed-Loop Motor Control, Life-Cycle Analysis, and Risk Assessment

Use the guidelines below to complete your report and add at the end of your report.

Group Number and Member Names:

Score	Pts	
	5	**General Format - Professional Looking Document/Preparation (whole document)** a)Fonts, margins (11pt, times new roman, single spaced. 1" margins on all sides). b)Spelling and grammar are correct c)Layout of pictures – all figures need numbers and captions and must be referenced inthe text d)Follows the page limitations below. e)References. Use IEEE reference format. f)This grading sheet is included as the final page.
	5	**Page 1: Title, Group Name, Group Members, and Date** **Executive Summary** (1 concise, well-written paragraph) Provide an overview of this project. Briefly describe what you did and what you learned.
		Control Subsystem Design
	5	**Page 2: Overview of Hardware-in-the-Loop** (~1/2 page) Describe in your own words what Hardware-in-the-Loop means. What is the difference between a full simulation and a Hardware-in-the-Loop simulation? What are the strengths of HIL?
	10	**Pages 2-3: Document Hardware** (1 page) Describe and show images of the equipment used, connection diagrams, calculation of resolution – pixels per square inch/cm on game board etc. Is the camera an appropriate sensor?
	10	**Pages 4-5: Document Software** (2 pages) Using Flowcharts, state diagrams, data structures etc. describe how the software is implemented. There is no need to include the source code.
	10	**Page 6: Document Motor Experiments** (~1 page) Plot the effect of changing gains using the reference, actual position and error signals. What happens when the proportional, derivative and/or integral gains are changed?
	15	**Pages 7: Document and Evaluate your User Interface** (~1 pages) How does the user interface connect the Camera-As-A-Sensor and the motor? What information is provided to the user and why? Document using screenshots and similar images.
	10	**Pages 8-9 Life Cycle Analysis (2 pages)** You are proposing a design that consumes resources. Follow the "Life Cycle Assessment (LCA) Exercise" for the shipping box for your project to examine the life cycle for this one part of your design. Be sure to interpret the results of the computer program. Complete this section of the report by saying that a similar analysis could be done on the entire project to reduce environmental impact.
	10	**Page 10-11: Financial Analysis** Provide a financial analysis that examines turning your group into a start-up company. Use the spreadsheet provided to make calculations and report your results using the MS Word document template and include here.
	15	**Page 12: Employee Training Program**
	15	**Pages 13-14: Safety Analysis** The project must be safe for use by the customer. Perform and Document a DFMEA for your project. Document your analysis using the DFMEA Table and Risk Assessment Matrix shown in the class slides. Show that you have implemented the results of the analysis to make your design and workspace safe – that is document what changes you made to make your system and space safe as a result of the safety analysis. Can you conclude your system is safe?
		Page 19: Grading Sheet